

El Moustpha Cheikh JIDDOU

Data Scientist – ML/Deep Learning Models for Forecasting, NLP and Computer Vision

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in El Moustpha Cheikh Jiddou | MoustaphaCheikhJidou | Portfolio |

PROFESSIONAL EXPERIENCE

Data Systems & IT Architecture Intern – Banque Nationale de Mauritanie (BNM)  *February 2025 – Present*

-  *Nouakchott, Mauritania*
- Implemented a RAG (Retrieval-Augmented Generation) architecture with PostgreSQL/pgvector to query official banking documentation without data leakage.
- Benchmarked several LLMs (GPT-4o, Claude 3.5, Gemini 1.5, local Llama) to optimize performance and confidentiality.
- Developed sequential neural network models (LSTM, GRU) to anticipate rapid photovoltaic power drops.

Assistant Engineer Intern – Mining & Geotechnical Department – Kinross Gold Corporation (Tasiast)
 *July 2025 – September 2025*

-  *Tasiast, Mauritania*
- PV short-term forecasting: prepared minute-level data (lags, ramps ΔP , time encodings) and prototyped sequential models (LSTM/GRU) to anticipate power drops.
- Environment:** Python (NumPy, pandas, Plotly, Gradio), Power BI, Git.

End-of-Study Internship – Caisse des Dépôts et de Développement (CDD)  *February – June 2023*

-  *Nouakchott, Mauritania*
- Used econometric models and modern machine learning approaches to predict default risks.
- Developed computational solutions for risk analysis (logistic regression, k-NN, Random Forest).

PROJECTS

Grain Size Analysis of 316L Microstructure – Computer-vision pipeline for grain boundary analysis

- Computer-vision pipeline segmenting grain boundaries in 316L stainless steel micrographs, comparing thresholding, gradient filters, HED, and a U-Net model. GitHub.

Disaster Tweets NLP Classification – Tweet classification using machine learning

- Developed a machine learning model to classify tweets related to natural disasters, using advanced NLP techniques. GitHub.

RAG-WebSystem – Generative AI system based on Retrieval-Augmented Generation

- End-to-end RAG engine: FAISS semantic search, contextual answer generation, exposed via FastAPI and React. GitHub.

EDUCATION

Higher Polytechnic School of Nouakchott – ESP - Mauritania  *October 2023 – June 2026*
State Engineer in Statistics and Data Engineering.

Faculty of Economic Sciences and Management – FSEG - Mauritania  *October 2023 – June 2025*
Master's in Applied Econometrics and Statistics.

SKILLS

PyTorch • TensorFlow • Keras • Python • Deep Learning • NLP • Computer Vision

LANGUAGES

Arabic 
Native Language

French 
DALF C1

English 
CEFR B1

VOLUNTEERING

President, ESP Data Club (January 2023 – April 2025)

CERTIFICATIONS

Google Advanced Data Analytics • DataCamp Data Science in Python • freeCodeCamp Data Analysis with Python